

PROJECT REPORT ON

**PLASTIC SHREDDER AND PLASTIC INJECTION
PROTOTYPE**

Submitted in partial fulfillment of the requirements for the award of the degree of

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MECHANICAL (AUTOMOBILE) ENGINEERING

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ABSTRACT

A plastic is any material that can be shaped or molded into any form -- some are naturally occurring, but most are man-made. Plastics are made from oil. Oil is a carbon-rich raw material, and plastics are large carbon-containing compounds. They're large molecules called polymers, which are composed of repeating units of shorter carbon-containing compounds called monomers. Most plastic is chemically inert and will not react chemically with other substances. Plastic can be molded into an almost infinite variety of shapes, so you can find it in toys, cups, bottles, utensils, wiring, cars, even in bubble gum. Plastics have revolutionized the world.

Due to their relatively low cost, ease of manufacture, versatility, and imperviousness to water, plastics are used in an enormous and expanding range of products, from paper clips to spaceships. They have already displaced many traditional materials, such as wood, stone, horn and bone, leather, paper, metal, glass, and ceramic, in most of their former uses. In developed countries, about a third of plastic is used in packaging and another third in buildings such as piping used in plumbing etc

Our project deals with reusing the plastic that's otherwise disposed to the environment and causes pollution. In our project we firstly shred the plastic in our plastic shredder and then with the help of our injection machine we have tried to create a new product that can be reused like a pen stand, paperweight etc.

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CHAPTER – 01

INTRODUCTION

A plastic is any material that can be shaped or molded into any form -- some are naturally occurring, but most are man-made. Plastics are made from oil. Oil is a carbon-rich raw material, and plastics are large carbon-containing compounds. They're large molecules called polymers, which are composed of repeating units of shorter carbon-containing compounds called monomers. Chemists combine various types of monomers in many different arrangements to make an almost infinite variety of plastics with different chemical properties. Most plastic is chemically inert and will not react chemically with other substances -- you can store alcohol, soap, water, acid or gasoline in a plastic container without dissolving the container itself. Plastic can be molded into an almost infinite variety of shapes, so you can find it in toys, cups, bottles, utensils, wiring, cars, even in bubble gum. Plastics have revolutionized the world.

Due to their relatively low cost, ease of manufacture, versatility, and imperviousness to water, plastics are used in an enormous and expanding range of products, from paper clips to spaceships. They have already displaced many traditional materials, such as wood, stone, horn and bone, leather, paper, metal, glass, and ceramic, in most of their former uses. In developed countries, about a third of plastic is used in packaging and another third in buildings such as piping used in plumbing etc

1.1 PLASTIC SHREDDING – OVERVIEW

Plastic shredding is the process by which the plastic is shredded into miniature form like flakes. The plastic is fed into the shredder that is made of blades connected to a shaft. This shaft is connected to a motor. The shredded plastic is collected and used in the injection machine or can be disposed of more efficiently.

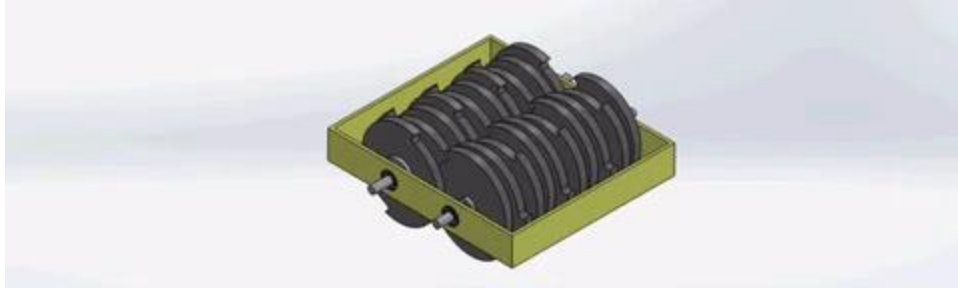


Fig. 1.1 Double shaft shredder blades.

1.2 INJECTION MOLDING-OVERVIEW:

Injection molding is a manufacturing process for producing parts from both thermoplastic and thermosetting plastic materials. Material is fed into a heated barrel, mixed, and forced into a mold cavity where it cools and hardens to the configuration of the mold cavity. Injection molding is widely used for manufacturing a variety of parts, from the smallest component to entire body panels of cars.

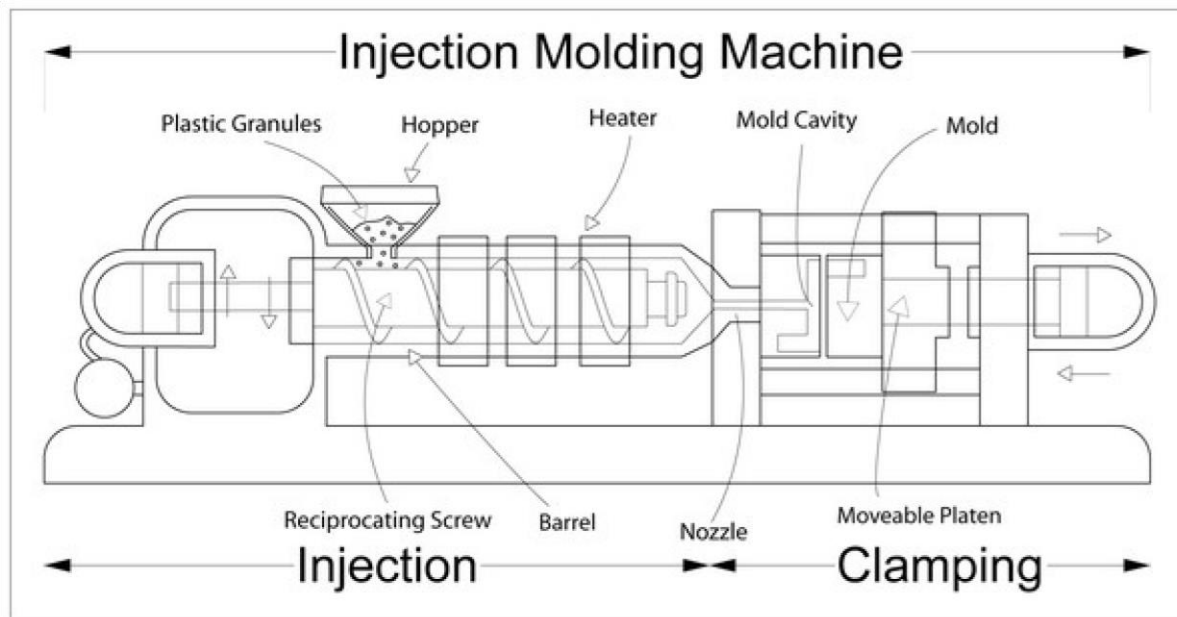


Fig. 1.2 Schematic Diagram of Plastic Injection molding

CHAPTER-2

LITERATURE SURVEY

1. Garrett C. Fitzgerald

Sustainable waste management of post-recycling municipal solid wastes (MSW) is an important component in the 'green' movement toward a cleaner, environmentally conscious society. Waste-to-Energy (WTE) power plants have potential to significantly reduce the amount of landfilled refuse while producing a carbon neutral form of heat and power, However, ; the average capital investment for a new WTE facility ranges from \$7,500 to \$9,000 per installed kW of capacity, nearly three times that of coal fire power plants. There exists a need to considerably reduce the cost of such facilities in order to bring them into the mainstream of solid waste management. This report examines how size reduction and homogenization of the raw MSW stream can potentially improve WTE operating characteristics while decreasing capital investments. Chemical rate and heat transfer theories indicate that the productivity of a moving grate WTE boiler should be enhanced by means of pre-shredding the MSW, thus reducing the average particle size, homogenizing the feed, and increasing its bulk density by an estimated 30%. Smaller particle sizes enhance reaction kinetics and flame propagation speed, due to the higher surface to volume ratio, and thus lower the amount of combustion air needed to meet the required combustion rates. Minimizing the primary combustion air supply rate lowers the total amount of flue gases and can result in decreased costs of the Air Pollution Control system. Smaller and more homogeneous particles increase bed mixing coefficients and reduce retention time required for complete combustion. The benefits realized through the pre-processing of MSW by means of modern shredding equipment were evaluated quantitatively both for the traditional High-Speed, Low-torque (HSLT) hammer mills and the new generation of LSHT shear shredders. The shearing mechanism utilized in these low rpm devices produce a more uniform particle distribution at a lower energy cost per ton MSW processed than hammer mills of the same capacity. The integration of size reduction systems into the typical flow sheet of WTE facilities has been hindered by the high frequency of fires, explosions, and ejected material from hammer mill grinders. The low shaft speed of the shear shredders has reduced the occurrence of fires and explosions while nearly eliminating ejected materials, allowing for safer and more reliable adaptation into new and existing WTE facilities.

The most important criterion in the adoption of pre-shredding MSW for grate combustion will require that economic and energy benefits of pre-shredding be clearly greater than the conventional operation of combusting as received MSW. At an average WTE electrical production of 650 kWh per metric ton of MSW processed, the required 3-11 kWh/ton for LSHT devices is less than 2% and should be more than accounted for by improved combustion efficiency in the WTE plant. The addition of a shredding system in a medium sized WTE plant will increase the O&M from current costs by roughly 10%, not including the benefit of lower maintenance due to improved distribution of thermal stresses on the grate and in the boiler. Finally, for the capital cost of a new WTE facility in the range of \$8000 per kW of capacity, the initial investment in shredding and fuel handling equipment will increase capital costs by about 2% from current values. It should be determined on a case-by-case basis whether the addition of pre-shredding equipment may increase capacity and decrease maintenance sufficiently to cover capital and operational costs as well as lower overall cost of operating the facility.

2. Lahiri K. K., Jena, Kapila K

All Blood Banks, Diagnostic Pathology and Microbiology Laboratories generate significant amounts of Bio-medical Waste (BMW), which poses a serious hazard to the clientele, community and environment at large. Microbiology waste (category 3) generated in the form of specimens, culture plates and tubes, stock cultures, antibiotic sensitivity plates etc., have the highest potential to spread infection and therefore they need the utmost care in their disposal by autoclaving/ chemical treatment/ microwaving or incineration. Amongst waste sharps the issue of 1% hypochlorite being inefficient in decontaminating blood containing hypodermic needles, autoclaving puncture proof needle containers before sending them to needle pits and capping of vacutainer needles, need to be reviewed. The other issues that will be discussed are incineration of Formalin fixed anatomical waste, since they release toxic gases and autoclaving of discarded/infected blood units in blood banks Liquid waste and effluent management need more serious deliberations. Segregation of waste at source and a sound scientific basis for appropriate disposal of laboratory generated bio-medical waste are the key steps in the management of waste in laboratory Medicine.

3. Vishal N. Kshirsagar

This paper describes about the experimentation of can or plastic bottle crusher machine and analysis of mechanism used in machine. Hence in this the knowledge of analysis is necessary, and by analysis of various parts the quality and life of machine can be increased and improved.

Overall, for experimentation this machine involves processes like design, fabrication, analysis and assembling of different components etc. From this the knowledge of all the parameters like design, fabrication and analysis etc. will get increase but most important the knowledge of analysis, the use of Ansys-Workbench Software is increasing day by day to determine the parameters like stress, strain, deflection etc. for safe design and long durability.

CHAPTER 3

SHREDDER MACHINE

Plastics are synthetic organic materials produced by polymerization. They are typically of high molecular mass, and may contain other substances besides polymers to improve performance and or reduce costs. These polymers can be moulded or extruded into desired shapes.

There are two main types of plastics first is “thermoplastics” and other one is “thermosetting” polymers. Thermoplastics can repeatedly soften and melt if enough heat is applied and hardened on cooling, so that they can be made into new plastics products. Examples are polyethylene, polystyrene and polyvinyl chloride, among others. Thermosets or thermosetting can melt and take shape only once. They are not suitable for repeated heat treatments; therefore after they have solidified, they stay solid. Examples are phenol formaldehyde and urea formaldehyde. Prior to their conversion into fuel resources, waste plastics are subject to various methods of pre-treatment to facilitate the smooth and efficient treatment during the subsequent conversion process. Depending on their structures (e.g. rigid, films, sheets or expanded (foamed) material) the pre-treatment equipment used for each type of plastic (crushing or shredding) is often different.

3.1. SHREDDER MODIFICATIONS

First generation of shredders: - Most of the first generation of the transmission mechanism is driven by a belt with low noise.

Second generation of shredders: - Plastic gear rolls, because it is difficult to master injection and shrinking process accurately of the shredder machine, resulting in the low accuracy of the gear itself.

Third generation shredders: - Metal sprocket: quiet operation, low energy loss, efficient cutting, and the perfect coordination of the various components of the system achieve the compelling features.

Fourth generation of shredder machine.- The drive mechanism of shredder machine is the metal gear, although the metal gear so overcome the above drawbacks, it is difficult to avoid the impact of the metal gear and friction sound.

Fifth generation of shredder:- Diamond snug movement, it takes use of alloy steel materials, quenching process of metal tool, completely CNC machining technology, and the workmanship guarantee transmission installation accuracy.

Sixth generation of shredders (modern):- Currently, the high-tech multimedia high series grinder has the high technology content which can be used to broken CD-ROM, floppy disk, tape, video, etc. and the embedded button panel with a protective film ensure the function of the way forward, rewind, stop, and full stop. In the modern world, we pay attention to care for the quality of life

3.2. WORKING

We have seen, plastic disposables, dirty and used plastics, plastic toys, bottles, containers and many others, which people stopped using and trashed due to breakage, old fashioned and many other reasons. Throwing them in a dustbin is not only a solution and it doesn't mean it can't be used further. As these plastics are reusable and can be easily mould, as per the way you like, thus, accumulating the same by big industries, helping to clean our environment as well as cutting down the expenses. Once these wastes have been accumulated, they directly go to these creative and efficient shredders and rest of the work it performs very well.

Shredders contain dual or many hex shafts and various knife designs, once it starts working, these shafts and knives reconfigured the plastic wastes in the desired output.

3.3. TYPES OF SHREDDING MACHINE

Many devices capable of material size reduction are available on the market ranging from automobile shredders, which are able to process almost anything, to granulators and paper shredders that can process only relatively soft materials. There are two prominent categories of shredders used in the management of MSW; high-speed, low-torque (HSLT) hammer mills and low-speed, high-torque (LSHT) shear shredders. Low torque shredders such as the top fed horizontal hammer mill utilize high speed rotating shafts (700-1200 rpm) that are equipped with fixed or pinned hammers used to crush the incoming material. The principal difference between these machines and the LSHT devices is that hammer mills rely almost entirely on impact and abrasive forces to smash the refuse into smaller particles. Figure 12 shows an axial cross section

of the rotating shaft and hammer, this drawing highlights the impact forces used in these machines to size reduce the refuse. It is important to notice that the hammer mills do not have tight tolerances between the hammers and cutting or sizing bars; this is because size reduction is primarily a result of the hammer smashing the MSW. Due to their reliance on impact force, hammer mills are generally more effective in processing brittle materials and can have problems with rags and stringy materials which can wrap around the shaft and cause overloading and disruption of the operation. The HSLT shredders have specific energy consumptions ranging from 6-22 kWh/ton depending on the characteristic size of the shredded refuse and the material composition. A study by Trezek on MSW size reduction has shown that the specific energy consumption of a hammer mill can be optimized by lowering the rotor speed by 25%. In this test, when the rotor speed was reduced from 1200 to 790 rpm, there was a 26 % reduction in power consumption for an equivalent amount of MSW processed on a per ton basis. The reason for this can be attributed to the fact that up to 20 % of a HSLT devices power is used to overcome bearing friction and windage of the rotor. If the machine is not loaded properly and consistently, a large fraction of the energy is used in idle spinning of the rotor. The speed of the rotor plays a significant role in rotor windage and internal pressure in the shredding compartment.

3.4. SIZE OF SHREDDER

The size and geometry of a shredder is quite important when developing an efficient integrated system. The goal of MSW size reduction is to increase productivity and decrease capital cost of a WTE facility. The footprint of such facilities will have significant ties to the overall cost of constructing a new plant. If shredders are to be effective in improving the WTE process they will need to be compact and smoothly integrated into the existing waste handling system

CHAPTER-04

SHREDDER COMPONENTS

4.1 MOTORS

4.1.1 AC motor

An industrial type of AC motor with electrical terminal box at the top and output rotating shaft on the left. Such motors are widely used for pumps, blowers, conveyors and other industrial machinery.

An AC motor is an electric motor driven by an alternating current (AC). The AC motor commonly consists of two basic parts, an outside stationary stator having coils supplied with alternating current to produce a rotating magnetic field, and an inside rotor attached to the output shaft producing a second rotating magnetic field. The rotor magnetic field may be produced by permanent magnets, reluctance saliency, or DC or AC electrical windings.



Fig. 4.1 AC Motor

Less commonly, linear AC motors operate on similar principles as rotating motors but have their stationary and moving parts arranged in a straight line configuration, producing linear motion instead of rotation.

4.1.1.1 Operating principles

When an AC motor is in steady-state rotation, the magnetic fields of the rotor and stator rotate with little or no slippage. The magnetic forces between the rotor and stator poles create average torque, capable of driving a load at rated speed. The speed of the stator rotating magnetic field and the speed of the rotor rotating magnetic field, relative to the speed of the mechanical shaft must maintain synchronism for average torque production by satisfying the synchronous speed relation. Otherwise, asynchronously rotating magnetic fields would produce pulsating or non-average torque.

The two main types of AC motors are classified as induction and synchronous. The induction motor (or asynchronous motor) always relies on a small difference in speed between the stator rotating magnetic field and the rotor shaft speed called slip to induce rotor current in the rotor AC winding. As a result, the induction motor cannot produce torque near synchronous speed where induction (or slip) is irrelevant or ceases to exist. In contrast, the synchronous motor does not rely on slip-induction for operation and uses either permanent magnets, salient poles (having projecting magnetic poles), or an independently excited rotor winding. The synchronous motor produces its rated torque at exactly synchronous speed. The brushless wound-rotor doubly fed synchronous motor system has an independently excited rotor winding that does not rely on the principles of slip-induction of current. The brushless wound-rotor doubly fed motor is a synchronous motor that can function exactly at the supply frequency or sub to super multiple of the supply frequency.

4.1.2 Induction motor

4.1.2.1 Slip

If the rotor of a squirrel cage motor were to run at the true synchronous speed, the flux in the rotor at any given place on the rotor would not change, and no current would be created in the squirrel cage. For this reason, ordinary squirrel-cage motors run at some tens of RPM slower than synchronous speed. Because the rotating field (or equivalent pulsating field) effectively rotates faster than the rotor, it could be said to slip past the surface of the rotor. The difference between synchronous speed and actual speed is called slip, and loading the motor increases the

amount of slip as the motor slows down slightly. Even with no load, internal mechanical losses prevent the slip from being zero.

The speed of the AC motor is determined primarily by the frequency of the AC supply and the number of poles in the stator winding, according to the relation:

$$N_s = 120F/p$$

where

N_s = Synchronous speed, in revolutions per minute

F = AC power frequency

p = Number of poles per phase winding

Actual RPM for an induction motor will be less than this calculated synchronous speed by an amount known as slip, that increases with the torque produced. With no load, the speed will be very close to synchronous. When loaded, standard motors have between 2-3% slip, special motors may have up to 7% slip, and a class of motors known as torque motors are rated to operate at 100% slip (0 RPM/full stall).

The slip of the AC motor is calculated by:

$$S = (N_s - N_r)/N_s$$

Where

N_r = Rotational speed, in revolutions per minute.

S = Normalised Slip, 0 to 1.

As an example, a typical four-pole motor running on 60 Hz might have a nameplate rating of 1725 RPM at full load, while its calculated speed is 1800 RPM.

The speed in this type of motor has traditionally been altered by having additional sets of coils or poles in the motor that can be switched on and off to change the speed of magnetic field rotation. However, developments in power electronics mean that the frequency of the power supply can also now be varied to provide a smoother control of the motor speed.

This kind of rotor is the basic hardware for induction regulators, which is an exception of the use of rotating magnetic field as pure electrical (not electromechanical) application.

4.1.3 Single-phase induction motor

Single-phase motors do not have a unique rotating magnetic field like multi-phase motors. The field alternates (reverses polarity) between pole pairs and can be viewed as two fields rotating in opposite directions. They require a secondary magnetic field that causes the rotor to move in a specific direction. After starting, the alternating stator field is in relative rotation with the rotor. Several methods are commonly used:.

4.1.4 Split-phase motor

Another common single-phase AC motor is the split-phase induction motor, commonly used in major appliances such as air conditioners and clothes dryers. Compared to the shaded pole motor, these motors provide much greater starting torque.

A split-phase motor has a secondary startup winding that is 90 electrical degrees to the main winding, always centered directly between the poles of the main winding, and connected to the main winding by a set of electrical contacts. The coils of this winding are wound with fewer turns of smaller wire than the main winding, so it has a lower inductance and higher resistance. The position of the winding creates a small phase shift between the flux of the main winding and the flux of the starting winding, causing the rotor to rotate. When the speed of the motor is sufficient to overcome the inertia of the load, the contacts are opened automatically by a centrifugal switch or electric relay. The direction of rotation is determined by the connection between the main winding and the start circuit. In applications where the motor requires a fixed rotation, one end of the start circuit is permanently connected to the main winding, with the contacts making the connection at the other end.

4.1.5 Synchronous motor

Three-phase system with rotating magnetic fields.

4.1.5.1 Polyphase synchronous motor

If connections to the rotor coils of a three-phase motor are taken out on slip-rings and fed a separate field current to create a continuous magnetic field (or if the rotor consists of a permanent magnet), the result is called a synchronous motor because the rotor will rotate synchronously with the rotating magnetic field produced by the polyphase electrical supply. Another synchronous motor system is the brushless wound-rotor doubly fed synchronous motor system with an independently excited rotor multiphase AC winding set that may experience slip-

induction beyond synchronous speeds but like all synchronous motors, does not rely on slip-induction for torque production.

Nowadays, synchronous motors are frequently driven by transistorized variable-frequency drives. This greatly eases the problem of starting the massive rotor of a large synchronous motor. They may also be started as induction motors using a squirrel-cage winding that shares the common rotor: once the motor reaches synchronous speed, no current is induced in the squirrel-cage winding so it has little effect on the synchronous operation of the motor, aside from stabilizing the motor speed on load changes.

Synchronous motors are occasionally used as traction motors; the TGV may be the best-known example of such use. Huge numbers of three phase synchronous motors are now fitted to electric cars. They have a Nd or other rare earth permanent magnet.

One use for this type of motor is its use in a power factor correction scheme. They are referred to as synchronous condensers. This exploits a feature of the machine where it consumes power at a leading power factor when its rotor is over excited. It thus appears to the supply to be a capacitor, and could thus be used to correct the lagging power factor that is usually presented to the electric supply by inductive loads. The excitation is adjusted until a near unity power factor is obtained (often automatically). Machines used for this purpose are easily identified as they have no shaft extensions. Synchronous motors are valued in any case because their power factor is much better than that of induction motors, making them preferred for very high power applications.

Some of the largest AC motors are pumped-storage hydroelectricity generators that are operated as synchronous motors to pump water to a reservoir at a higher elevation for later use to generate electricity using the same machinery. Six 500-megawatt generators are installed in the Bath County Pumped Storage Station in Virginia, USA. When pumping, each unit can produce 642,800 horsepower.

4.1.5.2 Single-phase synchronous motor

If a conventional squirrel-cage rotor has flats ground on it to create salient poles and increase reluctance, it will start conventionally, but will run synchronously, although it can provide only a modest torque at synchronous speed. This is known as a reluctance motor.

Because inertia makes it difficult to instantly accelerate the rotor from stopped to synchronous speed, these motors normally require some sort of special feature to get started.

Some include a squirrel-cage structure to bring the rotor close to synchronous speed. Various other designs use a small induction motor (which may share the same field coils and rotor as the synchronous motor) or a very light rotor with a one-way mechanism (to ensure that the rotor starts in the "forward" direction). In the latter instance, applying AC power creates chaotic (or seemingly chaotic) jumping movement back and forth; such a motor will always start, but lacking the anti-reversal mechanism, the direction it runs is unpredictable. The Hammond organ tone generator used a non-self-starting synchronous motor (until comparatively recently), and had an auxiliary conventional shaded-pole starting motor. A spring-loaded auxiliary manual starting switch connected power to this second motor for a few seconds.

4.2 V-BELT

V belts solved the slippage and alignment problem. It is now the basic belt for power transmission. They provide the best combination of traction, speed of movement, load of the bearings, and long service life. They are generally endless, and their general cross-section shape is trapezoidal (hence the name "V"). The "V" shape of the belt tracks in a mating groove in the pulley (or sheave), with the result that the belt cannot slip off. The belt also tends to wedge into the groove as the load increases—the greater the load, the greater the wedging action—improving torque transmission and making the V-belt an effective solution, needing less width and tension than flat belts. V-belts trump flat belts with their small center distances and high reduction ratios. The preferred center distance is larger than the largest pulley diameter, but less than three times the sum of both pulleys. Optimal speed range is 1,000–7,000 ft/min (300–2,130 m/min). V-belts need larger pulleys for their thicker cross-section than flat belts.

For high-power requirements, two or more V-belts can be joined side-by-side in an arrangement called a multi-V, running on matching multi-groove sheaves. This is known as a multiple-V-belt drive (or sometimes a "classical V-belt drive").

V-belts may be homogeneously rubber or polymer throughout, or there may be fibers embedded in the rubber or polymer for strength and reinforcement. The fibers may be of textile materials such as cotton, polyamide (such as Nylon) or polyester or, for greatest strength, of steel or aramid.

When an endless belt does not fit the need, jointed and link V-belts may be employed. Most models offer the same power and speed ratings as equivalently-sized endless belts and do not require special pulleys to operate. A link v-belt is a number of polyurethane/polyester composite links held together, either by themselves, such as Fenner Drives' PowerTwist, or by metal studs, such as Gates' Nu-T-Link. These provide easy installation and superior environmental resistance compared to rubber belts and are length adjustable by disassembling and removing links when needed.



Fig. 4.2 V- Belt

4.3 Hollow Pipe

A hollow structural section (HSS) is a type of metal profile with a hollow tubular cross section. HSS members can be circular, square, or rectangular sections, although other shapes are available, such as elliptical.

During the manufacturing process, flat steel plate is gradually changed in shape to become round where the edges are presented ready to weld. The edges are then welded together to form the mother tube. During the manufacturing process the mother tube goes through a series of shaping stands which form the round HSS (mother tube) into the final square or rectangular shape. HSS is only composed of structural steel per code.

Working Principle

The setup is made of hollow circular pipe of diameter 4 inches. Both ends of the hollow pipe is covered and a shaft of 12mm diameter with a blade setup is provided. The shaft is coupled to an electric motor which in turn rotates the shaft with blade. Bearings are provided at both ends for smooth rotation of the shaft. A hopper shape is provided over the steel pipe through which plastic particles are fed inside and a hole on the opposite side allows the small particles to move down.

CHAPTER-5

INJECTION MOULDING

5.1 PROCESS CYCLE:

The process cycle for injection molding is very short, typically between 2 seconds and 2 minutes, and consists of the following four stages:

1. **Clamping** - Prior to the injection of the material into the mold, the two halves of the mold must first be securely closed by the clamping unit. Each half of the mold is attached to the injection molding machine and one half is allowed to slide. The hydraulically powered clamping unit pushes the mold halves together and exerts sufficient force to keep the mold securely closed while the material is injected. The time required to close and clamp the mold is dependent upon the machine - larger machines (those with greater clamping forces) will require more time. This time can be estimated from the dry cycle time of the machine.
2. **Injection** - The raw plastic material, usually in the form of pellets, is fed into the injection molding machine, and advanced towards the mold by the injection unit. During this process, the material is melted by heat and pressure. The molten plastic is then injected into the mold very quickly and the buildup of pressure packs and holds the material. The amount of material that is injected is referred to as the shot. The injection time is difficult to calculate accurately due to the complex and changing flow of the molten plastic into the mold. However, the injection time can be estimated by the shot volume, injection pressure, and injection power.

3. **Cooling** - The molten plastic that is inside the mold begins to cool as soon as it makes contact with the interior mold surfaces. As the plastic cools, it will solidify into the shape of the desired part. However, during cooling some shrinkage of the part may occur. The packing of material in the injection stage allows additional material to flow into the mold and reduce the amount of visible shrinkage. The mold can not be opened until the required cooling time has elapsed. The cooling time can be estimated from several thermodynamic properties of the plastic and the maximum wall thickness of the part.
4. **Ejection** - After sufficient time has passed, the cooled part may be ejected from the mold by the ejection system, which is attached to the rear half of the mold. When the mold is opened, a mechanism is used to push the part out of the mold. Force must be applied to eject the part because during cooling the part shrinks and adheres to the mold. In order to facilitate the ejection of the part, a mold release agent can be sprayed onto the surfaces of the mold cavity prior to injection of the material. The time that is required to open the mold and eject the part can be estimated from the dry cycle time of the machine and should include time for the part to fall free of the mold. Once the part is ejected, the mold can be clamped shut for the next shot to be injected.

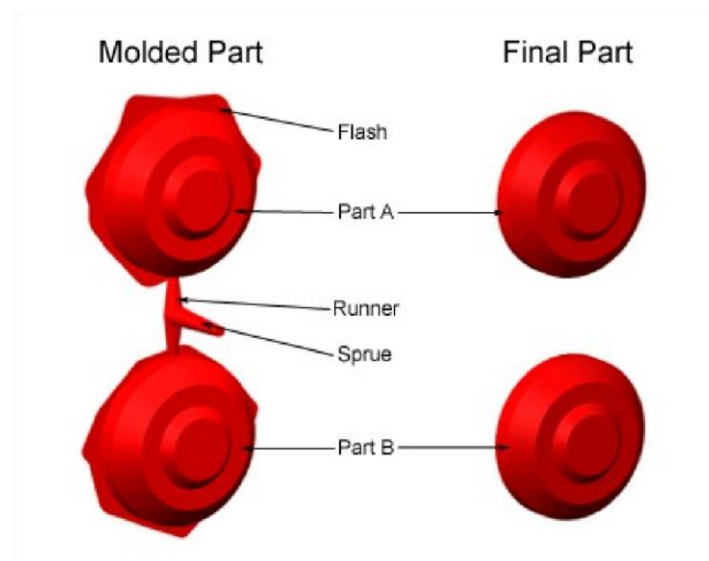


Fig.5.1 Injection molded part.

After the injection molding cycle, some post processing is typically required. During cooling, the material in the channels of the mold will solidify attached to the part. This excess material, along with any flash that has occurred, must be trimmed from the part, typically by using cutters. For some types of material, such as thermoplastics, the scrap material that results from this trimming can be recycled by being placed into a plastic grinder, also called regrind machines or granulators, which regrinds the scrap material into pellets. Due to some degradation of the material properties, the regrind must be mixed with raw material in the proper regrind ratio to be reused in the injection molding process.

5.2 MACHINERY & EQUIPMENT :

Injection molding machines consist of a material hopper, an injection ram or screw-type plunger, and a heating unit. They are also known as presses, they hold the molds in which the components are shaped. Presses are rated by tonnage, which expresses the amount of clamping force that the machine can exert. This force keeps the mold closed during the injection process. Tonnage can vary from less than 5 tons to 6000 tons, with the higher figures used in comparatively few manufacturing operations.

The total clamp force needed is determined by the projected area of the part being molded. This projected area is multiplied by a clamp force of from 2 to 8 tons for each square inch of the projected areas. As a rule of thumb, 4 or 5 tons/in² can be used for most products. If the plastic material is very stiff, it will require more injection pressure to fill the mold, thus more clamp tonnage to hold the mold closed. The required force can also be determined by the material used and the size of the part, larger parts require higher clamping force.

Injection molding machines have many components and are available in different configurations, including a horizontal configuration and a vertical configuration. However, regardless of their design, all injection molding machines utilize a power source, injection unit, mold assembly, and clamping unit to perform the four stages of the process cycle.

5.3 POWER REQUIREMENTS:

The power required for this process of injection molding depends on many things and varies between materials used. Manufacturing Processes Reference Guide states that the power requirements depend on "a material's specific gravity, melting point, thermal conductivity, part

size, and molding rate." Below is a table from page 243 of the same reference as previously mentioned which best illustrates the characteristics relevant to the power required for the most commonly used materials.

Material	Specific Gravity Melting Point (°F)	
Epoxy	1.12 to 1.24	248
Phenolic	1.34 to 1.95	248
Nylon	1.01 to 1.15	381 to 509
Polyethylene	0.91 to 0.965	230 to 243
Polystyrene	1.04 to 1.07	338

Table 1 Power Requirements.

5.4 INJECTION UNIT:

The injection unit is responsible for both heating and injecting the material into the mold. The first part of this unit is the hopper, a large container into which the raw plastic is poured. The hopper has an open bottom, which allows the material to feed into the barrel. The barrel contains the mechanism for heating and injecting the material into the mold. This mechanism is usually a ram injector or a reciprocating screw. A ram injector forces the material forward through a heated section with a ram or plunger that is usually hydraulically powered. Today, the more common technique is the use of a reciprocating screw. A reciprocating screw moves the material forward by both rotating and sliding axially, being powered by either a hydraulic or electric motor.

The material enters the grooves of the screw from the hopper and is advanced towards the mold as the screw rotates. While it is advanced, the material is melted by pressure, friction, and additional heaters that surround the reciprocating screw. The molten plastic is then injected very quickly into the mold through the nozzle at the end of the barrel by the buildup of pressure and the forward action of the screw. This increasing pressure allows the material to be packed and forcibly held in the mold. Once the material has solidified inside the mold, the screw can retract and fill with more material for the next shot.

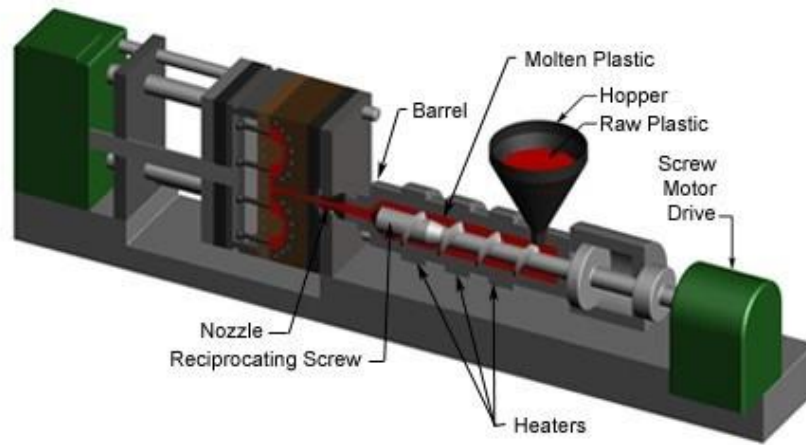


Fig.5.2 Injection molding machine - Injection unit.

5.5 CLAMPING UNIT:

Prior to the injection of the molten plastic into the mold, the two halves of the mold must first be securely closed by the clamping unit. When the mold is attached to the injection molding machine, each half is fixed to a large plate, called a platen. The front half of the mold, called the mold cavity, is mounted to a stationary platen and aligns with the nozzle of the injection unit. The rear half of the mold, called the mold core, is mounted to a movable platen, which slides along the tie bars. The hydraulically powered clamping motor actuates clamping bars that push the moveable platen towards the stationary platen and exert sufficient force to keep the mold securely closed while the material is injected and subsequently cools. After the required cooling time, the mold is then opened by the clamping motor. An ejection system, which is attached to the rear half of the mold, is actuated by the ejector bar and pushes the solidified part out of the open cavity.

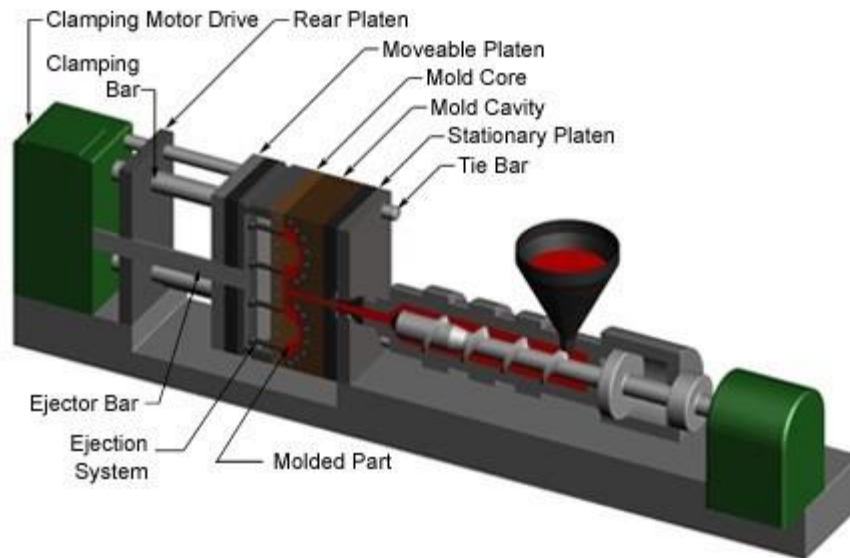


Fig.5.3 Injection molding machine - Clamping unit.

5.6 LUBRICATION AND COOLING:

Obviously, the mold must be cooled in order for the production to take place. Because of the heat capacity, inexpensiveness, and availability of water, water is used as the primary cooling agent. To cool the mold, water can be channeled through the mold to account for quick cooling times. Usually a colder mold is more efficient because this allows for faster cycle times. However, this is not always true because crystalline materials require the opposite: a warmer mold and lengthier cycle time.

5.7 TOOLING:

The injection molding process uses molds, typically made of steel or aluminum, as the custom tooling. The mold has many components, but can be split into two halves. Each half is attached inside the injection molding machine and the rear half is allowed to slide so that the mold can be opened and closed along the mold's parting line. The two main components of the mold are the mold core and the mold cavity. When the mold is closed, the space between the mold core and the mold cavity forms the part cavity, that will be filled with molten plastic to

create the desired part. Multiple-cavity molds are sometimes used, in which the two mold halves form several identical part cavities.

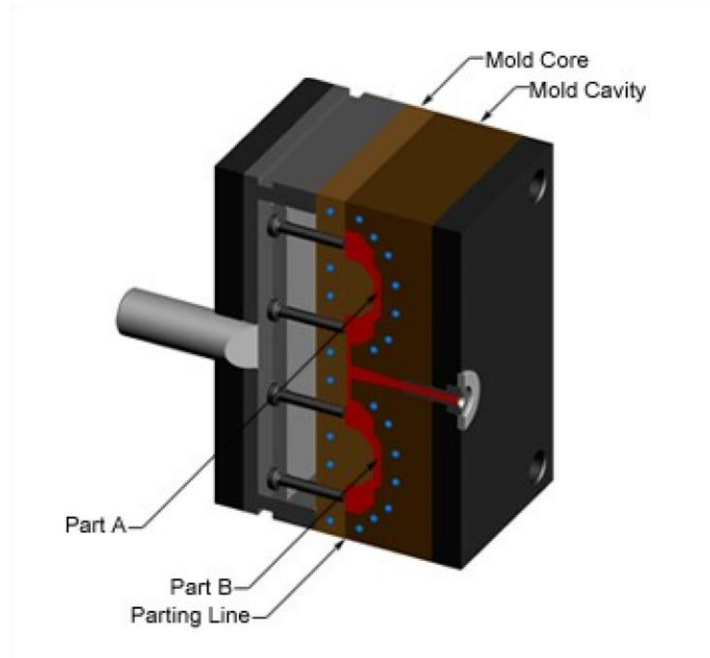


Fig.5.4 Mold overview.

5.8 MOULD BASE:

The mold core and mold cavity are each mounted to the mold base, which is then fixed to the platens inside the injection molding machine. The front half of the mold base includes a support plate, to which the mold cavity is attached, the sprue bushing, into which the material will flow from the nozzle, and a locating ring, in order to align the mold base with the nozzle. The rear half of the mold base includes the ejection system, to which the mold core is attached, and a support plate. When the clamping unit separates the mold halves, the ejector bar actuates the ejection system. The ejector bar pushes the ejector plate forward inside the ejector box, which in turn pushes the ejector pins into the molded part. The ejector pins push the solidified part out of the open mold cavity.

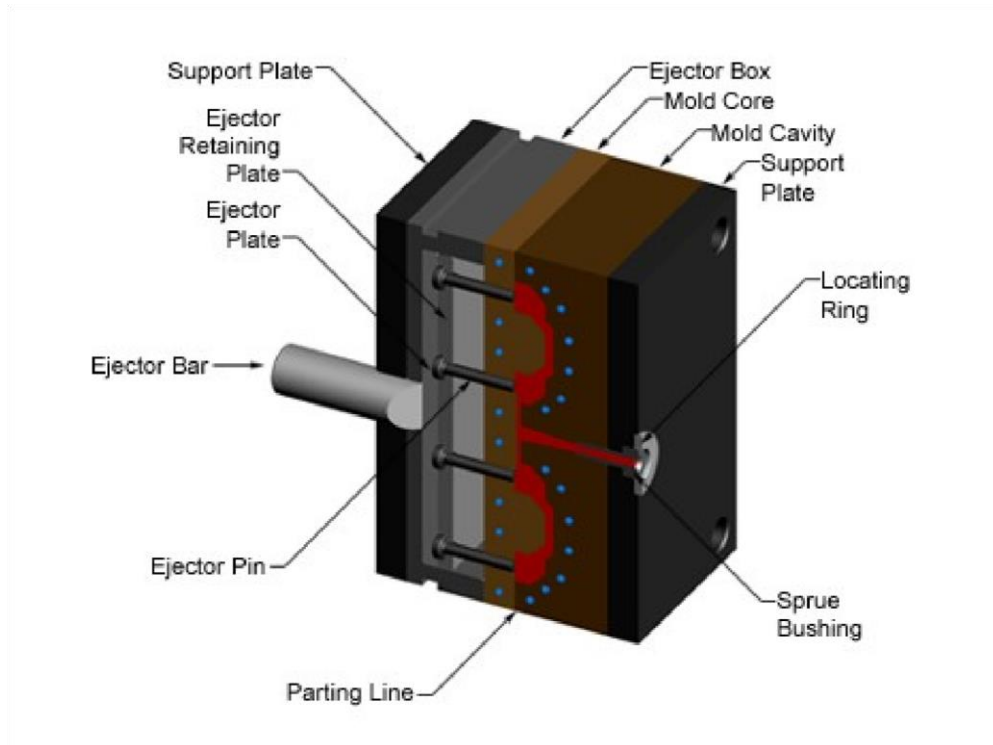


Fig.5.5 Mold base.

5.9 MOULD CHANNELS:

In order for the molten plastic to flow into the mold cavities, several channels are integrated into the mold design. First, the molten plastic enters the mold through the sprue. Additional channels, called runners, carry the molten plastic from the sprue to all of the cavities that must be filled. At the end of each runner, the molten plastic enters the cavity through a gate which directs the flow. The molten plastic that solidifies inside these runners is attached to the part and must be separated after the part has been ejected from the mold. However, sometimes hot runner systems are used which independently heat the channels, allowing the contained material to be melted and detached from the part. Another type of channel that is built into the mold is cooling channels. These channels allow water to flow through the mold walls, adjacent to the cavity, and cool the molten plastic.

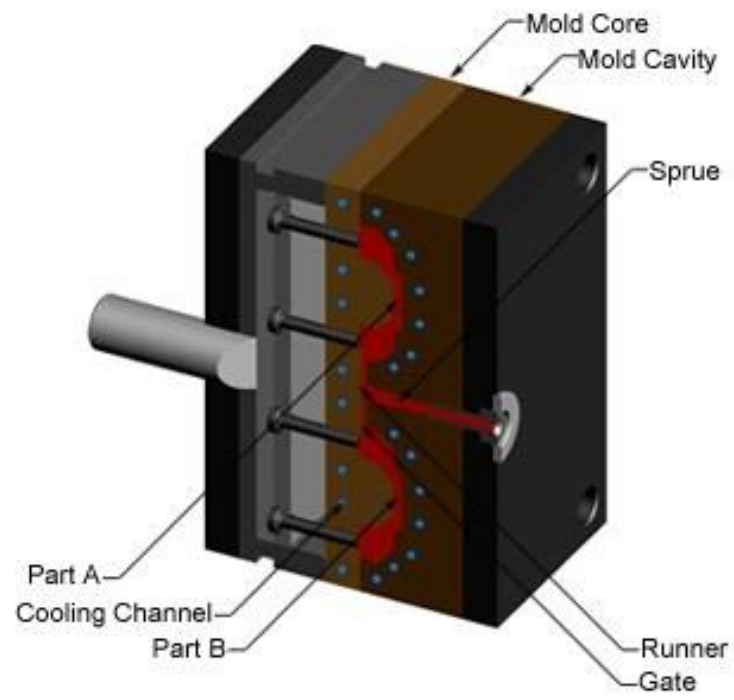


Fig.5.6 Mold channels.

CHAPTER- 6

DESIGNS

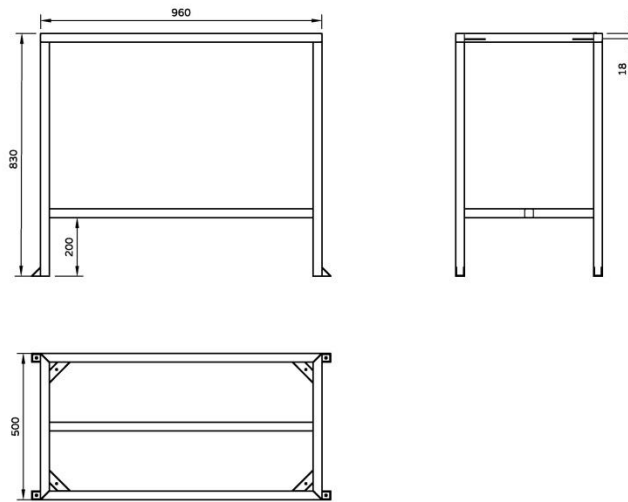


Fig.6.1 Shredder Framework

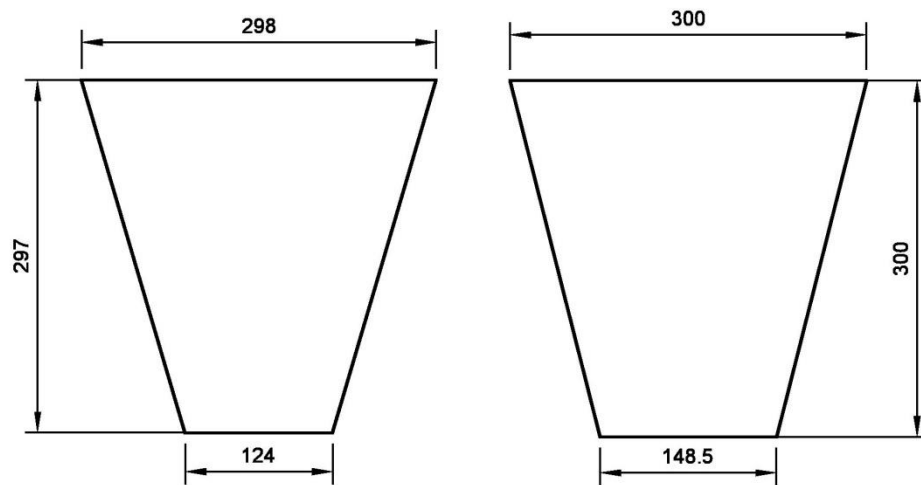


Fig.6.2 Hopper Sheets

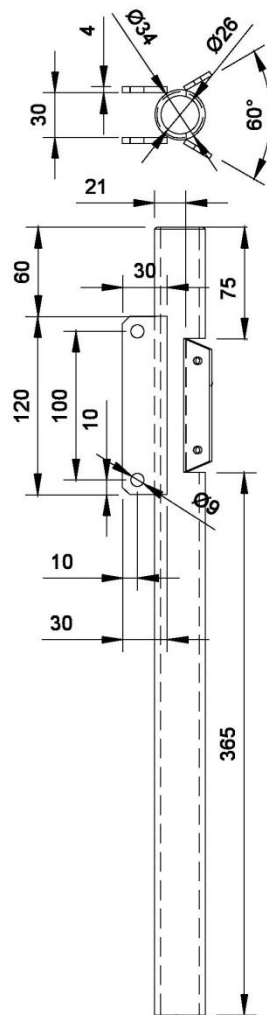


Fig.6.3 Injection Barrel

CHAPTER- 7

APPLICATIONS

Injection molding is used to create many things such as wire spools, packaging, bottle caps, automotive dashboards, pocket combs, and most other plastic products available today. Injection molding is the most common method of part manufacturing. It is ideal for producing high volumes of the same object. Some advantages of injection molding are high production rates, repeatable high tolerances, and the ability to use a wide range of materials, low labor cost, minimal scrap losses, and little need to finish parts after molding. Some disadvantages of this process are expensive equipment investment, potentially high running costs, and the need to design moldable parts.

Most polymers may be used, including all thermoplastics, some thermo sets, and some elastomers. In 1995 there were approximately 18,000 different materials available for injection molding and that number was increasing at an average rate of 750 per year. The available materials are alloys or blends of previously developed materials meaning that product designers can choose from a vast selection of materials, one that has exactly the right properties. Materials are chosen based on the strength and function required for the final part but also each material has different parameters for molding that must be taken into account.^[8] Common polymers like Epoxy and phenolic are examples of thermosetting plastics while nylon, polyethylene, and polystyrene are thermoplastic.

7.1 GENERAL APPLICATIONS:

- Aerospace components
- Automotive components
- Avionics components
- Cable assemblies
- Computer electronics
- Electronics components
- Encapsulations
- Engineering prototypes
- Geophysics

- Instrumentation
- Marketing samples
- Material quality testing
- Medical & dental products
- Medical laboratories
- Model shops, toys, hobby
- New product design & development
- R&D labs
- Test specimens

7.2 FUTURE

Some of the new tendencies and technology in injection molding are the electric injection machines and the gas assisted injection molding. The electric machines have several advantages over the old design of the conventional injection machine. It runs silent, its operating cost is less, and they are more accurate and stable.



Fig.7.1 An all-electrical Injection Machine.

CHAPTER-08

CONCLUSION

The plastic shredding machine is widely used in industries for the plastic waste management. By using this recycling plastic shredding machine, the overall difficulty of the shredding process gets reduced. It requires less labor work and there is no requirement of any skilled labor. The main problem with our shredder is that for cutting most of the plastics that are available in our household, it requires blades which are thicker and precision cut which is not economical for everyone. The shredder also requires a more powerful motor running at low speeds.

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